Statistics: Questions and Answers

Stages A to G, one short answer per topic

A1. Understanding data

Q1. What is statistics? Descriptive vs. inferential statistics.

A. Descriptive statistics summarizes data; inferential statistics draws conclusions about a population from a sample.

Q2. Population, sample, observation, variable, and parameter.

A. Population is the whole group, sample is a subset, observation is one record, variable is a measured attribute, and parameter is a true population value.

Q3. Qualitative vs. quantitative data.

A. Qualitative data describes categories; quantitative data describes numeric amounts.

Q4. Nominal, ordinal, interval, and ratio measurement scales.

A. Nominal is unordered labels, ordinal is ordered labels, interval has equal gaps but no true zero, and ratio has equal gaps and a true zero.

Q5. Discrete vs. continuous variables.

A. Discrete variables take countable values; continuous variables take any value in a range.

Q6. Independent variables, dependent variables, features, and targets.

A. Independent variables are inputs, the dependent variable is the outcome, features are ML inputs, and the target is the ML output.

Q7. Structured vs. unstructured data.

A. Structured data has defined columns like a table; unstructured data is text, images, or audio.

Q8. Data collection, data sources, and data quality.

A. Collection is how data is gathered, sources are where it comes from, and quality is how accurate, complete, and consistent it is.

Q9. Missing values, duplicates, invalid values, and inconsistent categories.

A. Missing values are blanks, duplicates are repeated rows, invalid values break rules (like attendance of 120%), and inconsistent categories are spelling variants like "AI/ML" and "ai-ml".

Q10. Observational data vs. experimental data.

A. In observational data the researcher only watches; in experimental data the researcher assigns the treatment.

A2. Measures of central tendency

Q11. Mean (arithmetic average).

A. The mean is the sum divided by the count, and it is sensitive to outliers.

Q12. Median.

A. The median is the middle value when sorted, and it is robust to outliers.

Q13. Mode.

A. The mode is the most frequent value and is the only option for categories.

Q14. Weighted mean.

A. The weighted mean gives each value a weight, such as course grades weighted by credits.

Q15. Geometric mean.

A. The geometric mean is the nth root of the product of n values, used for growth rates.

Q16. Harmonic mean.

A. The harmonic mean is the reciprocal of the mean of reciprocals, used for averaging rates like speed.

Q17. Trimmed mean.

A. The trimmed mean drops a set percentage from both ends before averaging, which resists outliers.

Q18. Choosing the appropriate measure for a dataset.

A. Use the median for skewed data, the mean for symmetric data, the mode for categories, and weighted means when groups differ in importance.

A3. Measures of dispersion (spread)

Q19. Minimum and maximum.

A. Minimum and maximum are the smallest and largest values.

Q20. Range.

A. The range is the maximum minus the minimum.

Q21. Variance (population and sample).

A. Variance is the average squared deviation from the mean, dividing by n for population and n-1 for sample.

Q22. Standard deviation (population and sample).

A. Standard deviation is the square root of variance, in the original units.

Q23. Interquartile range (IQR).

A. The IQR is Q3 minus Q1, the spread of the middle 50%.

Q24. Quartiles.

A. Quartiles are the values that split sorted data into four equal parts.

Q25. Mean absolute deviation.

A. Mean absolute deviation is the average absolute distance from the mean.

Q26. Coefficient of variation.

A. The coefficient of variation is the standard deviation divided by the mean, a unitless relative spread.

Q27. Absolute vs. relative variability.

A. Absolute variability is in original units, while relative variability lets you compare spread across different scales.

A4. Measures of position and standardized values

Q28. Quartiles, deciles, and percentiles.

A. Quartiles split data into four parts, deciles into ten, and percentiles into a hundred.

Q29. Five-number summary.

A. The five-number summary is min, Q1, median, Q3, and max, which is what a boxplot draws.

Q30. Z-scores and standardization.

A. A z-score is (x - mean) / std, the distance from the mean in standard deviations.

Q31. Ranking and relative standing.

A. Ranking shows where a value sits relative to all others in the dataset.

Q32. Empirical rule (68-95-99.7).

A. In normal data about 68%, 95%, and 99.7% of values fall within 1, 2, and 3 standard deviations of the mean.

Q33. Percentile ranks.

A. A percentile rank is the percentage of values at or below a given value.

A5. Shape of a distribution

Q34. Symmetric and asymmetric distributions.

A. A symmetric distribution has mirror-image halves, while an asymmetric one has one longer tail.

Q35. Positive (right) skewness.

A. Positive (right) skew has a long right tail, with the mean usually above the median.

Q36. Negative (left) skewness.

A. Negative (left) skew has a long left tail, with the mean usually below the median.

Q37. Kurtosis: tails and outlier-proneness.

A. Kurtosis measures tail weight: positive excess kurtosis means heavier tails than normal, negative means lighter.

Q38. Unimodal, bimodal, and multimodal distributions.

A. Unimodal has one peak, bimodal has two (possibly two hidden groups), and multimodal has several.

Q39. Floor and ceiling effects.

A. Floor and ceiling effects occur when values pile up at a minimum or maximum imposed by the measurement.

Q40. Outliers and extreme observations.

A. Outliers are values far from the rest, which should be checked for errors before any decision.

A6. Frequency and categorical analysis

Q41. Frequency tables.

A. A frequency table counts how often each category appears.

Q42. Relative and cumulative frequency.

A. Relative frequency is each count's share of the total; cumulative frequency is the running total.

Q43. Grouped frequency distributions.

A. A grouped frequency distribution counts values within ranges of a numeric variable.

Q44. Cross-tabulation and contingency tables.

A. A cross-tabulation (contingency table) counts combinations of two categorical variables.

Q45. Row percentages vs. column percentages.

A. Row percentages divide by row totals and column percentages divide by column totals, so choose based on your question.

Q46. Counts vs. proportions.

A. Counts depend on group size, while proportions allow fair comparison between groups.

Q47. Category imbalance and rare categories.

A. Rare categories with very few rows produce unstable results and may need merging.

A7. Data visualization

Q48. Histograms.

A. A histogram shows how values of one numeric variable are distributed across bins.

Q49. Bar charts and stacked bar charts.

A. A bar chart shows counts per category; a stacked bar chart also shows parts within each bar.

Q50. Box plots.

A. A box plot shows the median, quartiles, and outliers.

Q51. Violin plots.

A. A violin plot combines a box plot with a density curve to show shape.

Q52. Density plots.

A. A density plot is a smooth curve estimating the distribution's shape.

Q53. Dot plots and stem-and-leaf plots.

A. A dot plot shows each value as a point, and a stem-and-leaf plot keeps the raw digits visible.

Q54. Scatter plots.

A. A scatter plot shows the relationship between two numeric variables.

Q55. Line charts and time-series plots.

A. Line charts and time-series plots show values across time or in order.

Q56. ECDF (empirical cumulative distribution function).

A. An ECDF shows the share of data at or below each value, without binning.

Q57. Q-Q plots for comparing distributions.

A. A Q-Q plot compares sample quantiles to theoretical ones; a straight line suggests a matching distribution.

Q58. Choosing a chart based on variable type.

A. Choose histograms or box plots for numeric data, bar charts for categories, and scatter plots for two numeric variables.

Q59. Misleading axes, bin choices, and visualization pitfalls.

A. Truncated axes, poor bin choices, and 3D effects can distort what a chart shows.

A8. Outlier detection

Q60. IQR rule (1.5 x IQR).

A. The IQR rule flags values below Q1 - 1.5 x IQR or above Q3 + 1.5 x IQR.

Q61. Z-score method.

A. The z-score method flags values whose absolute z-score exceeds a cutoff, usually 3.

Q62. Modified Z-score using median and MAD.

A. The modified z-score uses the median and MAD, so it works better for skewed data.

Q63. Univariate vs. multivariate outliers.

A. Univariate outliers are extreme in one column; multivariate outliers are unusual in combination with other columns.

Q64. Data errors vs. legitimate extreme values.

A. Data errors should be fixed or removed, while legitimate extremes are usually kept.

Q65. Winsorization, capping, transformation, and removal.

A. Winsorization caps values at chosen percentiles, transformation compresses them, and removal deletes the rows.

Q66. Why outliers should be investigated before being removed.

A. Investigate outliers first, because they can be errors or valuable signal.

A9. Relationships between variables

Q67. Covariance.

A. Covariance measures how two variables move together in their original units, and its sign gives the direction.

Q68. Pearson correlation.

A. Pearson correlation measures linear association on a scale from -1 to +1.

Q69. Spearman rank correlation.

A. Spearman correlation is the correlation of ranks and captures monotonic relationships.

Q70. Kendall rank correlation.

A. Kendall correlation is based on concordant and discordant pairs and is robust for small or tied samples.

Q71. Positive, negative, and zero correlation.

A. Positive correlation means both rise together, negative means one rises as the other falls, and zero means no linear pattern.

Q72. Linear vs. nonlinear relationships.

A. A linear relationship follows a straight line, while a nonlinear one curves, and Pearson can miss curved patterns.

Q73. Correlation vs. causation.

A. Correlation does not show that one variable causes the other.

Q74. Confounding variables and spurious correlations.

A. A confounder influences both variables, and a spurious correlation appears without any real link.

Q75. Simpson's paradox.

A. Simpson's paradox is when a trend in combined data reverses within each subgroup.

Q76. Multicollinearity (introductory concept).

A. Multicollinearity is when predictors are highly correlated with each other, which makes coefficients unstable.

B1. Basic probability concepts

Q77. Random experiments.

A. A random experiment is a process with an uncertain result, such as checking one student's placement.

Q78. Outcomes and sample spaces.

A. The sample space is the set of all possible outcomes.

Q79. Events and event notation.

A. An event is a subset of outcomes, such as "placed".

Q80. Certain, impossible, and complementary events.

A. A certain event has probability 1, an impossible one has 0, and the complement is everything not in the event.

Q81. Probability bounds: 0 to 1.

A. All probabilities lie between 0 and 1 inclusive.

Q82. Theoretical, empirical, and subjective probability.

A. Theoretical probability comes from logic, empirical from observed frequencies, and subjective from belief.

Q83. Equally likely outcomes.

A. Equally likely outcomes each have the same chance, like the sides of a fair die.

B2. Counting techniques

Q84. Fundamental counting principle.

A. The fundamental counting principle multiplies the number of options at each stage.

Q85. Factorials.

A. n! (factorial) is the product of 1 through n, the number of ways to order n items.

Q86. Permutations.

A. A permutation is an ordered selection of r items from n.

Q87. Combinations.

A. A combination is an unordered selection of r items from n.

Q88. With-replacement vs. without-replacement sampling.

A. With replacement, items go back before the next draw; without replacement, they don't.

Q89. When order matters and when it does not.

A. Order matters for rankings and arrangements, but not for committees or teams.

B3. Rules of probability

Q90. Addition rule.

A. The addition rule is P(A or B) = P(A) + P(B) - P(A and B).

Q91. Multiplication rule.

A. The multiplication rule is P(A and B) = P(A) x P(B|A).

Q92. Complement rule.

A. The complement rule is P(not A) = 1 - P(A).

Q93. Union and intersection of events.

A. A union is "either event happens", and an intersection is "both happen".

Q94. Mutually exclusive events.

A. Mutually exclusive events cannot both occur at once.

Q95. Independent vs. dependent events.

A. Independent events don't affect each other, while dependent events do.

Q96. Conditional probability.

A. Conditional probability P(A|B) = P(A and B) / P(B) is the probability of A given B.

Q97. Law of total probability.

A. The law of total probability combines conditional probabilities across a set of exhaustive cases.

Q98. Bayes' theorem.

A. Bayes' theorem P(A|B) = P(B|A)P(A) / P(B) reverses the direction of a conditional.

Q99. Base rates and the base-rate fallacy.

A. The base-rate fallacy is ignoring how rare a condition is when judging a positive result.

B4. Expected value and uncertainty

Q100. Random variables (discrete and continuous).

A. A random variable assigns numbers to outcomes and is either discrete or continuous.

Q101. Expected value (mathematical expectation).

A. Expected value is the long-run average of outcomes, weighted by their probabilities.

Q102. Variance of a random variable.

A. Variance of a random variable is the expected squared deviation from its expected value.

Q103. Standard deviation of a random variable.

A. Its standard deviation is the square root of that variance.

Q104. Linearity of expectation.

A. Linearity of expectation says E[X + Y] = E[X] + E[Y], even when X and Y are dependent.

Q105. Covariance and dependence between random variables.

A. Covariance measures how two random variables move together.

Q106. Conditional expectation.

A. Conditional expectation is the expected value of Y given a value of X.

Q107. Expected loss and expected utility (introductory).

A. Expected loss averages the cost of decisions under uncertainty, and expected utility weighs outcomes by their value.

B5. Probability applications

Q108. Placement probability.

A. Placement probability estimates the chance a student gets placed.

Q109. Probability of placement given internship experience.

A. Compare P(placed | internship) with P(placed | no internship) to see if internships matter.

Q110. Probability of multiple successes.

A. Probability of multiple successes counts how likely a given number of successes is across trials.

Q111. False positives and false negatives.

A. A false positive is a wrong flag, and a false negative is a missed true case.

Q112. Medical-test and fraud-detection examples.

A. A test's real-world accuracy depends on prevalence, so rare conditions yield many false positives.

Q113. Bayesian reasoning and updating beliefs.

A. Bayesian reasoning starts with a prior belief and updates it as new evidence arrives.

C1. Distribution fundamentals

Q114. Probability mass functions (PMF).

A. A PMF gives the probability of each value of a discrete variable.

Q115. Probability density functions (PDF).
A. A PDF gives the density for a continuous variable, where areas under the curve are probabilities.

Q116. Cumulative distribution functions (CDF).
A. A CDF gives the probability that a value is at or below x.

Q117. Discrete vs. continuous distributions.
A. Discrete variables take countable values, while continuous variables take any value in a range.

Q118. Parameters of a distribution.
A. Parameters such as the mean and variance define the shape and location of a distribution.

Q119. Mean, variance, and standard deviation of a distribution.
A. The mean is the center, and variance and standard deviation measure spread.

Q120. Joint, marginal, and conditional distributions.
A. A joint distribution covers two variables together, a marginal covers one, and a conditional covers one given the other.

Q121. Independent and identically distributed (i.i.d.) observations.
A. i.i.d. means observations are independent and drawn from the same distribution.

C2. Discrete distributions

Q122. Bernoulli distribution (one yes/no outcome).
A. Bernoulli models one yes/no trial, such as one student being placed.

Q123. Binomial distribution (number of successes).
A. Binomial counts successes in n independent yes/no trials.

Q124. Geometric distribution (trials until first success).
A. Geometric counts trials until the first success.

Q125. Negative binomial distribution.
A. Negative binomial counts trials until a set number of successes.

Q126. Hypergeometric distribution (sampling without replacement).
A. Hypergeometric counts successes when sampling without replacement from a finite group.

Q127. Poisson distribution (event counts).
A. Poisson counts events in a fixed time or space at a constant rate.

Q128. Discrete uniform distribution.
A. Discrete uniform gives every outcome the same probability, like a fair die.

C3. Continuous distributions

Q129. Continuous uniform distribution.
A. Continuous uniform gives every value in an interval equal density.

Q130. Normal (Gaussian) distribution.
A. Normal is the symmetric bell curve defined by its mean and standard deviation.

Q131. Standard normal distribution.
A. Standard normal is the normal distribution with mean 0 and standard deviation 1.

Q132. Exponential distribution.
A. Exponential models waiting times between events occurring at a constant rate.

Q133. Gamma distribution.
A. Gamma is a flexible positive-valued distribution for waiting times and sums of exponentials.

Q134. Beta distribution.
A. Beta models proportions between 0 and 1.

Q135. Weibull distribution.
A. Weibull models lifetimes and failure times where the hazard rate changes over time.

Q136. Log-normal distribution.

A. Log-normal means the log of the variable is normal, common for incomes and other right-skewed positives.

Q137. Student's t-distribution.

A. Student's t has heavier tails than normal and is used for means from small samples.

Q138. Chi-square distribution.

A. Chi-square is the distribution of a sum of squared standard normals, used for variance and categorical tests.

Q139. F-distribution.

A. F is the ratio of two chi-square variables, used in ANOVA and variance comparisons.

C4. Sampling distributions and theorems

Q140. Population distribution vs. sample distribution.

A. The population distribution covers all values, while the sample distribution covers the observed values.

Q141. Sampling distribution of the mean.

A. The sampling distribution of the mean is how sample means vary across repeated samples.

Q142. Sampling distribution of a proportion.

A. The sampling distribution of a proportion is how sample proportions vary across repeated samples.

Q143. Standard error.

A. Standard error is the standard deviation of a sampling distribution, equal to sigma / sqrt(n).

Q144. Central Limit Theorem (CLT).

A. The CLT says averages of many independent observations look approximately normal, regardless of the original shape.

Q145. Law of Large Numbers.

A. The Law of Large Numbers says the sample average converges to the population mean as the sample grows.

Q146. Normal approximation to the binomial.

A. The normal approximation to the binomial works when n is large and p is not extreme.

Q147. Relationship between sample size and sampling variability.

A. Larger samples reduce sampling variability, so estimates become more precise.

Q148. When the CLT does and does not justify a normal approximation.

A. The CLT needs independence and finite variance, and it can fail for very heavy tails or very small samples.

C5. Distribution assessment

Q149. Comparing observed data with theoretical distributions.

A. Compare observed data with a theoretical curve by overlaying it on a histogram or comparing quantiles.

Q150. Normality assessment.

A. Normality assessment checks whether data could plausibly come from a normal distribution.

Q151. Q-Q plots.

A. A Q-Q plot compares sample quantiles to normal quantiles, and a straight line suggests normality.

Q152. Shapiro-Wilk test (introductory).

A. The Shapiro-Wilk test checks normality, with a small p-value suggesting the data is not normal.

Q153. Probability calculations from a CDF.

A. Probabilities from a CDF give the chance a value falls below a given point.

Q154. Quantiles and inverse CDFs.

A. Quantiles and inverse CDFs give the value at a given probability, such as the 90th percentile.

Q155. Distribution fitting and goodness of fit.

A. Distribution fitting picks a distribution for the data, and goodness-of-fit tests check how well it matches.

D1. Sampling methods

Q156. Population and sampling frame.
A. The population is the full target group, and the sampling frame is the list you draw from.

Q157. Simple random sampling.
A. Simple random sampling gives every member an equal chance of selection.

Q158. Stratified sampling.
A. Stratified sampling splits the population into groups and samples within each one.

Q159. Cluster sampling.
A. Cluster sampling randomly selects whole groups and surveys people within them.

Q160. Systematic sampling.
A. Systematic sampling selects every kth member after a random start.

Q161. Multistage sampling.
A. Multistage sampling combines several sampling methods in successive stages.

Q162. Convenience and voluntary-response sampling.
A. Convenience and voluntary-response sampling use easily reached people and are usually biased.

Q163. Sampling bias and selection bias.
A. Sampling bias and selection bias are systematic errors in how the sample is chosen.

Q164. Nonresponse bias.
A. Nonresponse bias occurs when people who don't respond differ systematically from those who do.

Q165. Survivorship bias.
A. Survivorship bias happens when only the survivors are studied, ignoring those who dropped out or failed.

Q166. Sample size and representativeness.
A. Sample size reduces random error, but representativeness matters more than size.

Q167. Sampling with vs. without replacement.
A. Sampling with replacement can select the same item twice, while without replacement cannot.

D2. Estimation

Q168. Point estimates.
A. A point estimate is a single best guess for a population parameter.

Q169. Estimators vs. estimates.
A. An estimator is the rule for calculating a value, and an estimate is the number it produces.

Q170. Bias and unbiasedness.
A. Bias is systematic error, and an unbiased estimator has an expected value equal to the true parameter.

Q171. Consistency and efficiency.
A. A consistent estimator approaches the truth as n grows, and an efficient one has the smallest variance among comparable estimators.

Q172. Standard errors.
A. The standard error measures how precise an estimator is.

Q173. Margin of error.
A. The margin of error is the +/- part of a confidence interval, typically the critical value times the standard error.

Q174. Confidence intervals for means.
A. A confidence interval for a mean is the sample mean plus or minus the margin of error.

Q175. Confidence intervals for proportions.
A. A confidence interval for a proportion gives a plausible range for the true population proportion.

Q176. Confidence intervals for differences between means and proportions.
A. A confidence interval for a difference shows a plausible range for the gap between two groups.

Q177. Confidence intervals for correlation and regression coefficients.

A. Intervals for correlation or regression coefficients show a plausible range for the strength or slope of a relationship.

Q178. Bootstrap confidence intervals.

A. A bootstrap interval is built by repeatedly resampling the data with replacement.

Q179. Interpretation of a 95% confidence interval.

A. A 95% interval means that across repeated sampling, about 95% of such intervals would capture the true value.

D3. Hypothesis testing

Q180. Null and alternative hypotheses.

A. The null hypothesis H0 states no effect, and the alternative H1 states the effect you are testing for.

Q181. One-sided vs. two-sided tests.

A. A one-sided test checks one direction of difference, while a two-sided test checks for any difference.

Q182. Significance level (alpha).

A. The significance level alpha is the threshold for rejecting H0, commonly 0.05.

Q183. Test statistics.

A. A test statistic summarizes how far the data is from what H0 predicts.

Q184. P-values.

A. A p-value is the probability of data this extreme if H0 is true.

Q185. Critical values and rejection regions.

A. Critical values define the rejection region, the boundary beyond which H0 is rejected.

Q186. Statistical significance vs. practical significance.

A. Statistical significance means the result is unlikely by chance, while practical significance means it matters in real terms.

Q187. Type I error (false positive).

A. A Type I error is a false positive: rejecting H0 when it is true.

Q188. Type II error (false negative).

A. A Type II error is a false negative: failing to reject H0 when H1 is true.

Q189. Statistical power.

A. Statistical power is the probability of detecting a real effect, equal to 1 - beta.

Q190. Sample size and power calculations.

A. Power calculations determine the sample size needed to detect an effect of a given size.

Q191. Multiple testing and the risk of false discoveries.

A. Testing many hypotheses raises the chance of false discoveries, so you need corrections.

Q192. Pre-registration and the problem of p-hacking.

A. Pre-registration means stating your hypotheses and methods in advance to avoid p-hacking.

Q193. Why a non-significant result does not prove there is no effect.

A. A non-significant result doesn't prove no effect, because the study may lack power.

D4. Resampling methods

Q194. Bootstrap sampling.

A. Bootstrap sampling resamples the data with replacement to estimate variability.

Q195. Permutation tests.

A. A permutation test shuffles group labels to build a null distribution.

Q196. Randomization tests.

A. A randomization test checks whether an observed pattern could arise by chance through reshuffling.

Q197. Monte Carlo simulation.

A. Monte Carlo simulation uses repeated random draws to approximate a distribution or quantity.

Q198. Cross-validation (introduced here, developed further in Stage G).

A. Cross-validation splits data into folds to estimate performance on unseen data.

Q199. Confidence intervals and uncertainty from resampling.

A. Resampling gives uncertainty and confidence intervals without strong distributional assumptions.

E1. Comparing means

Q200. One-sample t-test.

A. A one-sample t-test compares one sample mean to a known value.

Q201. Independent two-sample t-test.

A. An independent two-sample t-test compares the means of two unrelated groups.

Q202. Welch's t-test.

A. Welch's t-test compares two independent means without assuming equal variances.

Q203. Paired-sample t-test.

A. A paired-sample t-test compares two measurements taken on the same subjects.

Q204. One-way ANOVA.

A. One-way ANOVA compares means across three or more groups.

Q205. Two-way ANOVA.

A. Two-way ANOVA tests the effects of two factors and their interaction.

Q206. Repeated-measures ANOVA.

A. Repeated-measures ANOVA compares means across repeated measurements on the same subjects.

Q207. Post-hoc tests: Tukey HSD and pairwise comparisons.

A. Post-hoc tests, like Tukey HSD, identify which specific group pairs differ after ANOVA.

Q208. Multiple comparisons and family-wise error.

A. The family-wise error rate is the chance of at least one false positive across many comparisons, and corrections control it.

Q209. Effect sizes for group comparisons.

A. Effect sizes for group comparisons, such as Cohen's d or eta-squared, measure how large the difference is.

E2. Categorical data tests

Q210. Chi-square goodness-of-fit test.

A. The chi-square goodness-of-fit test checks whether observed category counts match expected proportions.

Q211. Chi-square test of independence.

A. The chi-square test of independence checks whether two categorical variables are associated.

Q212. Chi-square test of homogeneity.

A. The chi-square test of homogeneity checks whether category distributions are the same across groups.

Q213. Fisher's exact test.

A. Fisher's exact test handles small counts in tables, especially 2x2.

Q214. McNemar's test for paired categorical outcomes.

A. McNemar's test compares paired yes/no outcomes, such as before and after.

Q215. Two-proportion z-test.

A. The two-proportion z-test compares two proportions using a normal approximation.

Q216. Tests for differences in rates and proportions.

A. Tests for rates and proportions compare how often an outcome occurs across groups.

Q217. Expected counts and sparse contingency tables.

A. Chi-square needs expected counts of about 5 or more per cell, and sparse tables call for Fisher's test.

E3. Non-parametric tests

Q218. Mann-Whitney U test.

A. The Mann-Whitney U test compares two independent groups using ranks, as an alternative to the t-test.

Q219. Wilcoxon signed-rank test.

A. The Wilcoxon signed-rank test compares paired differences using ranks.

Q220. Kruskal-Wallis test.

A. The Kruskal-Wallis test compares three or more groups using ranks, as an alternative to ANOVA.

Q221. Friedman test.

A. The Friedman test compares repeated measures across conditions using ranks.

Q222. Spearman correlation test.

A. The Spearman correlation test checks for a monotonic relationship using ranks.

Q223. Kendall correlation test.

A. The Kendall correlation test checks rank association using concordant and discordant pairs.

Q224. Sign test.

A. The sign test checks whether the median difference is zero by counting signs.

Q225. Kolmogorov-Smirnov test.

A. The Kolmogorov-Smirnov test compares a sample distribution to a reference distribution or to another sample.

Q226. Permutation-based alternatives to classical tests.

A. Permutation alternatives build a null distribution by reshuffling the data.

E4. Correlation and association tests

Q227. Pearson correlation significance test.

A. The Pearson correlation significance test checks whether r differs from zero.

Q228. Spearman rank correlation test.

A. The Spearman rank correlation test checks whether the rank correlation differs from zero.

Q229. Kendall rank correlation test.

A. The Kendall rank correlation test measures rank association through concordant and discordant pairs.

Q230. Point-biserial correlation.

A. Point-biserial correlation measures the association between a continuous and a binary variable.

Q231. Phi coefficient.

A. The phi coefficient measures association between two binary variables.

Q232. Cramer's V.

A. Cramer's V measures association strength for categorical tables of any size.

Q233. Partial correlation.

A. Partial correlation measures the relationship between two variables after removing the effect of others.

Q234. Correlation confidence intervals.

A. A correlation confidence interval gives the range of plausible values for the true correlation.

Q235. Multiple comparisons when testing many correlations.

A. Multiple-comparison corrections are needed when testing many correlations.

E5. Assumptions and diagnostics

Q236. Independence of observations.

A. Independence means one observation must not influence another, which repeated measures or clustered data can violate.

Q237. Normality of residuals or relevant sampling distributions.

A. Normality means residuals or sampling distributions should be approximately normal.

Q238. Equal variance (homoscedasticity).

A. Equal variance (homoscedasticity) means spread should be similar across groups or predictions.

Q239. Linearity.

A. Linearity means the relationship should be a straight line for linear methods.

Q240. Appropriate measurement level.

A. Appropriate measurement level means the test must match the variable type, such as interval versus nominal.

Q241. Outlier and influential-observation checks.

A. Influential-observation checks look for points that disproportionately drive the results.

Q242. Sphericity in repeated-measures ANOVA.

A. Sphericity in repeated-measures ANOVA means the variances of differences between conditions should be equal.

Q243. Expected-frequency assumptions in chi-square tests.

A. Expected-frequency assumptions in chi-square require enough observations in each cell.

Q244. Robust statistical methods.

A. Robust methods, such as trimmed or rank-based approaches, are less sensitive to assumption violations.

Q245. What to do when assumptions are violated.

A. When assumptions fail, you can transform the data, use a non-parametric test, or use a robust method.

E6. Experimental design and causal reasoning

Q246. Observational studies vs. experiments.

A. Observational studies observe what happens, while experiments assign the treatment.

Q247. Randomized controlled experiments.

A. In a randomized controlled experiment, the researcher randomly assigns treatment and measures outcomes.

Q248. Random assignment vs. random sampling.

A. Random assignment makes groups comparable, while random sampling makes results generalizable.

Q249. Control groups and treatment groups.

A. The treatment group receives the intervention, and the control group does not.

Q250. Placebo effects and blinding.

A. A placebo is an inactive treatment, and blinding hides group assignment to reduce bias.

Q251. Replication and reproducibility.

A. Replication means repeating a study to confirm its findings, and reproducibility means others can get the same results from the same data and methods.

Q252. Blocking and matched-pair designs.

A. Blocking groups similar subjects before randomizing reduces variation, and matched-pair designs compare matched individuals.

Q253. Factorial experiments and interaction effects.

A. Factorial experiments test several factors and their interactions at once.

Q254. Confounding and lurking variables.

A. Confounding and lurking variables are unmeasured or related factors that distort the apparent relationship.

Q255. Selection bias and measurement bias.

A. Selection bias and measurement bias are systematic errors from who is included or how outcomes are measured.

Q256. Internal vs. external validity.

A. Internal validity means the causal conclusion is correct, and external validity means it generalizes.

Q257. A/B testing.

A. A/B testing randomly compares two versions, such as two website designs, to measure which performs better.

Q258. Introduction to causal inference.

A. Causal inference uses design and assumptions to estimate cause-and-effect from data.

Q259. Correlation vs. causal effects.

A. Correlation shows association, while a causal effect requires a design or assumptions that rule out alternative explanations.

E7. Effect size and interpretation

Q260. Cohen's d.

A. Cohen's d is the standardized mean difference, the gap divided by the pooled standard deviation.

Q261. Hedges' g.

A. Hedges' g is Cohen's d corrected for small-sample bias.

Q262. Eta-squared and partial eta-squared.

A. Eta-squared is the share of variance explained by a factor, and partial eta-squared removes the effect of other factors.

Q263. Odds ratios.

A. An odds ratio compares the odds of an outcome between two groups.

Q264. Risk ratios and risk differences.

A. A risk ratio divides two probabilities, and a risk difference subtracts them.

Q265. Cramer's V.

A. Cramer's V measures association strength for categorical tables.

Q266. Confidence intervals around effect sizes.

A. A confidence interval around an effect size gives the range of plausible effect values.

Q267. Statistical power and minimum detectable effect.

A. Statistical power is the chance of detecting an effect, and the minimum detectable effect is the smallest effect a study can reliably find.

Q268. Practical significance and domain context.

A. Practical significance asks whether the effect matters in real-world terms, judged with domain knowledge.

F1. Simple linear regression

Q269. Dependent and independent variables.

A. The dependent variable is the outcome being predicted, and the independent variable is the predictor that explains it.

Q270. Regression line and fitted values.

A. The regression line is the line of best fit, and fitted values are the predictions on that line.

Q271. Intercept and slope.

A. The intercept is the predicted value when x is zero, and the slope is the change in y per unit change in x.

Q272. Least-squares estimation.

A. Least squares chooses the line that minimizes the sum of squared errors.

Q273. Residuals and residual sum of squares.

A. Residuals are observed minus predicted values, and the residual sum of squares adds up their squares.

Q274. R-squared and adjusted R-squared.

A. R-squared is the share of variance explained, and adjusted R-squared penalizes adding predictors that don't help.

Q275. Standard error of regression.

A. The standard error of regression is the typical size of residuals, in the units of y.

Q276. Confidence intervals and prediction intervals.

A. A confidence interval covers the mean prediction, and a prediction interval covers an individual value and is wider.

Q277. Significance tests for coefficients.

A. Significance tests for coefficients check whether each coefficient differs from zero.

Q278. Interpretation of the slope and intercept.

A. The slope is the change per unit of x, and the intercept is the baseline value, often not meaningful on its own.

F2. Multiple linear regression

Q279. Multiple predictors.

A. Multiple predictors explain the outcome together.

Q280. Categorical predictors and dummy variables.

A. Dummy variables convert categories into 0/1 columns.

Q281. Reference categories.

A. The reference category is the omitted category that the others are compared against.

Q282. Interaction terms.

A. An interaction term means the effect of one predictor depends on the level of another.

Q283. Polynomial regression.

A. Polynomial regression adds squared or higher-order terms to capture curves.

Q284. Nonlinear relationships.

A. Nonlinear relationships are patterns that don't follow a straight line and need transformations or nonlinear terms.

Q285. Adjusted comparisons between groups.

A. Adjusted comparisons between groups hold other variables constant.

Q286. Partial regression coefficients.

A. Partial regression coefficients give the effect of one predictor while holding the others fixed.

Q287. Multicollinearity and VIF.

A. Multicollinearity is high correlation among predictors, and VIF measures how much it inflates coefficient variance.

Q288. Model specification and omitted-variable bias.

A. Omitted-variable bias occurs when leaving out important variables distorts the coefficients of the ones included.

F3. Regression assumptions and diagnostics

Q289. Linearity.

A. Linearity means the relationship between predictors and the outcome has the right form.

Q290. Independence of errors.

A. Independence of errors means residuals are not correlated with each other, which matters in time-series or clustered data.

Q291. Constant variance of residuals.

A. Constant variance means residual spread is the same across fitted values.

Q292. Residual normality for classical small-sample inference.

A. Residual normality matters for small-sample inference on coefficients.

Q293. Multicollinearity.

A. Multicollinearity makes coefficients unstable and hard to interpret.

Q294. Residual plots.

A. Residual plots show residuals against fitted values to spot patterns.

Q295. Q-Q plots of residuals.

A. Q-Q plots of residuals compare their quantiles to a normal distribution.

Q296. Leverage and influential observations.

A. Leverage measures how extreme a point's predictor values are.

Q297. Cook's distance.

A. Cook's distance measures how much removing a point changes the model.

Q298. Heteroscedasticity tests.

A. Heteroscedasticity tests check whether residual variance changes with the predictors.

Q299. Robust standard errors.

A. Robust standard errors adjust inference when the variance is not constant.

Q300. Train/test evaluation vs. in-sample fit.

A. Train/test evaluation judges the model on held-out data, while in-sample fit only reflects the training data.

F4. Logistic regression

Q301. Binary outcome variables.

A. A binary outcome has two categories, such as placed or not placed.

Q302. Odds and probability.

A. Odds are P / (1 - P), while probability is P itself.

Q303. Odds ratios.

A. An odds ratio compares the odds between groups or per unit change in a predictor.

Q304. Log-odds and the logit function.

A. The logit is the log of the odds, and logistic regression models it as a linear function of predictors.

Q305. Logistic regression coefficients.

A. A logistic coefficient is the change in log-odds per unit change in the predictor.

Q306. Maximum likelihood estimation.

A. Maximum likelihood estimation finds the coefficients that make the observed outcomes most probable.

Q307. Predicted probabilities.

A. Predicted probabilities give the chance of the positive outcome for each individual.

Q308. Classification thresholds.

A. A classification threshold is the probability cutoff that turns a prediction into a yes or no decision.

Q309. Confounding and adjusted odds ratios.

A. Adjusted odds ratios measure the effect of a predictor while controlling for other variables.

Q310. Multinomial logistic regression.

A. Multinomial logistic regression handles more than two unordered categories.

Q311. Ordinal logistic regression.

A. Ordinal logistic regression handles ordered categories, such as low, medium, and high.

Q312. Model calibration.

A. Calibration means predicted probabilities match observed frequencies.

F5. Other regression models

Q313. Poisson regression for count outcomes.

A. Poisson regression models count outcomes.

Q314. Negative binomial regression for overdispersed counts.
A. Negative binomial regression handles count data where variance exceeds the mean.

Q315. Probit regression.
A. Probit regression is like logistic regression but uses the normal distribution as the link.

Q316. Quantile regression.
A. Quantile regression models conditional quantiles, such as the median, instead of the mean.

Q317. Ridge regression.
A. Ridge regression shrinks coefficients toward zero with a penalty to reduce overfitting.

Q318. Lasso regression.
A. Lasso regression uses a penalty that can shrink some coefficients exactly to zero, selecting features.

Q319. Elastic Net.
A. Elastic Net combines ridge and lasso penalties.

Q320. Generalized linear models (GLMs).
A. Generalized linear models unify linear, logistic, and Poisson models through a link function.

Q321. Generalized additive models (GAMs).
A. Generalized additive models use smooth flexible functions of predictors to capture nonlinear effects.

Q322. Mixed-effects and multilevel regression.
A. Mixed-effects models handle grouped data, such as students nested within colleges.

Q323. Survival analysis and time-to-event models.
A. Survival analysis models the time until an event, handling censored observations.

Q324. Cox proportional hazards model.
A. The Cox proportional hazards model estimates how predictors affect the hazard, or event rate, over time.

F6. Regression evaluation

Q325. MAE (mean absolute error).
A. MAE is the average absolute difference between predictions and actual values.

Q326. MSE (mean squared error).
A. MSE is the average squared error, which penalizes large errors more heavily.

Q327. RMSE (root mean squared error).
A. RMSE is the square root of MSE, in the original units of the outcome.

Q328. R-squared vs. predictive accuracy.
A. R-squared measures fit on the training data, while predictive accuracy measures performance on new data.

Q329. Residual analysis.
A. Residual analysis examines residuals for patterns that signal a problem with the model.

Q330. Out-of-sample evaluation.
A. Out-of-sample evaluation tests the model on data it did not see during training.

Q331. Prediction uncertainty.
A. Prediction uncertainty is the range of plausible predictions, not just one number.

Q332. Extrapolation and its risks.
A. Extrapolation means predicting outside the observed data range, which is unreliable.

Q333. Model interpretability.
A. Model interpretability is how easily people can understand the model's logic.

Q334. Prediction vs. causal explanation.
A. Prediction finds accurate outputs, while causal explanation requires assumptions about why the relationship exists.

G1. Data preparation and statistical preprocessing

Q335. Missing-data mechanisms: MCAR, MAR, and MNAR.

A. MCAR means missingness is random, MAR depends on observed data, and MNAR depends on the missing value itself.

Q336. Missing-value imputation: mean, median, mode, and model-based methods.

A. Missing-value imputation fills gaps with mean, median, mode, or a model-based estimate.

Q337. Complete-case analysis and its limitations.

A. Complete-case analysis drops incomplete rows, which loses data and can bias results.

Q338. Duplicate records and data leakage.

A. Duplicate records inflate the data, and leakage occurs when information from the test set enters training.

Q339. Feature scaling and standardization.

A. Standardization rescales features to mean 0 and standard deviation 1.

Q340. Min-max normalization.

A. Min-max normalization rescales features to a fixed range, usually 0 to 1.

Q341. Log, square-root, and Box-Cox transformations.

A. Log, square-root, and Box-Cox transformations reduce skew, with Box-Cox choosing the best power.

Q342. Encoding categorical variables.

A. Categorical encoding converts categories to numbers, such as one-hot encoding.

Q343. Handling rare categories.

A. Rare categories can be grouped together to avoid unstable estimates.

Q344. Class imbalance and resampling.

A. Class imbalance is handled with resampling or class weights.

Q345. Robust preprocessing in the presence of outliers.

A. Robust preprocessing uses methods like the median and robust scaling, which are less sensitive to outliers.

Q346. Preventing preprocessing leakage between training and test data.

A. Preprocessing leakage is avoided by fitting imputers and scalers on training data only, then applying them to test data.

G2. Machine Learning evaluation statistics

Q347. Training, validation, and test sets.

A. Training sets fit the model, validation sets tune it, and test sets give the final score.

Q348. Cross-validation and stratified cross-validation.

A. Cross-validation rotates folds so each serves as validation, and stratified versions keep class ratios.

Q349. Overfitting and underfitting.

A. Overfitting means the model memorizes noise, and underfitting means it misses real patterns.

Q350. Bias-variance trade-off.

A. The bias-variance trade-off balances simple models with high bias against complex models with high variance.

Q351. Confusion matrix.

A. A confusion matrix is a table of true versus predicted classes.

Q352. Accuracy, precision, recall, and specificity.

A. Accuracy is overall correctness, precision is how many positive predictions are right, recall is how many true positives are found, and specificity is how many negatives are correctly identified.

Q353. F1-score.

A. The F1-score is the harmonic mean of precision and recall.

Q354. ROC curve and AUC.

A. The ROC curve plots true positive rate against false positive rate, and AUC summarizes its area as ranking ability.

Q355. Precision-recall curve and average precision.

A. The precision-recall curve is especially useful for rare positive cases, and average precision summarizes it.

Q356. Threshold selection.

A. Threshold selection chooses the probability cutoff that balances the errors you care about.

Q357. Probability calibration and Brier score.

A. Probability calibration checks that predicted probabilities match observed frequencies, and the Brier score measures squared probability error.

Q358. Cost-sensitive evaluation.

A. Cost-sensitive evaluation assigns different costs to false positives and false negatives.

Q359. Baseline models.

A. A baseline model is a simple reference, such as always predicting the majority class.

Q360. Generalization and data leakage.

A. Data leakage inflates scores, while generalization means performing well on new data.

Q361. Model comparison and uncertainty in evaluation metrics.

A. Model comparison uses fold-to-fold variability and statistical tests to decide if one model is really better.

G3. Multivariate statistics

Q362. Covariance matrices.

A. A covariance matrix holds the variances and covariances among all features.

Q363. Correlation matrices.

A. A correlation matrix holds the pairwise correlations among features.

Q364. Partial correlation.

A. Partial correlation measures the relationship between two variables while controlling for others.

Q365. Principal Component Analysis (PCA).

A. PCA finds the directions of greatest variance to compress data into fewer components.

Q366. Singular Value Decomposition (SVD).

A. SVD is a matrix factorization that underlies PCA and many other methods.

Q367. Factor analysis.

A. Factor analysis explains correlations among variables through a smaller number of hidden factors.

Q368. Cluster analysis.

A. Cluster analysis groups similar observations together without labels.

Q369. Mahalanobis distance.

A. Mahalanobis distance measures distance from a center while accounting for correlations and scales.

Q370. Multivariate normal distribution.

A. The multivariate normal distribution generalizes the bell curve to several variables.

Q371. MANOVA.

A. MANOVA tests whether groups differ across several outcomes at once.

Q372. Discriminant analysis.

A. Discriminant analysis finds linear combinations of features that best separate classes.

Q373. Dimensionality reduction.

A. Dimensionality reduction cuts the number of features while keeping most of the information.

Q374. Multiple testing correction: Bonferroni and false discovery rate (FDR).

A. Bonferroni controls the family-wise error rate, and FDR controls the expected share of false discoveries.

G4. Bayesian statistics

Q375. Prior, likelihood, and posterior distributions.

A. The prior is your belief before seeing data, the likelihood is how probable the data is, and the posterior is the updated belief.

Q376. Bayesian updating.

A. Bayesian updating uses the current posterior as the prior when new data arrives.

Q377. Conjugate priors.

A. Conjugate priors belong to the same family as the posterior, which makes the math simple.

Q378. Credible intervals vs. confidence intervals.

A. A credible interval gives the probability the parameter lies inside it, while a confidence interval describes the procedure's long-run coverage.

Q379. Bayesian hypothesis testing.

A. Bayesian hypothesis testing compares hypotheses by their posterior probabilities.

Q380. Bayes factors.

A. A Bayes factor is the ratio of evidence for one model over another.

Q381. Posterior predictive distributions.

A. A posterior predictive distribution predicts new data while accounting for parameter uncertainty.

Q382. Hierarchical Bayesian models.

A. Hierarchical Bayesian models have group-level parameters that share information across groups.

Q383. Markov Chain Monte Carlo (MCMC).

A. MCMC draws samples from a posterior distribution when it can't be computed directly.

Q384. Bayesian regression.

A. Bayesian regression places priors on coefficients to get full uncertainty estimates.

G5. Time-series statistics

Q385. Time-series components: trend, seasonality, and noise.

A. Time-series components are trend, seasonality, and noise.

Q386. Autocorrelation and partial autocorrelation.

A. Autocorrelation measures how a series correlates with its lagged values, and partial autocorrelation removes the effect of intermediate lags.

Q387. Stationarity.

A. Stationarity means the mean and variance stay constant over time.

Q388. Differencing.

A. Differencing subtracts previous values to remove trend and make a series stationary.

Q389. Moving averages and exponential smoothing.

A. Moving averages and exponential smoothing average recent observations, with exponential weights favoring the latest data.

Q390. Autoregressive (AR) models.

A. An AR model predicts a value from its own past values.

Q391. Moving-average (MA) models.

A. An MA model predicts a value from past forecast errors.

Q392. ARIMA and seasonal ARIMA.

A. ARIMA combines autoregression, differencing, and moving averages, and seasonal ARIMA adds seasonal terms.

Q393. Forecast intervals.

A. Forecast intervals give the range expected to contain future values.

Q394. Time-based train/test splits.

A. Time-based splits train on earlier periods and test on later ones, never shuffling randomly.

Q395. Forecast accuracy metrics.

A. Forecast accuracy metrics like MAE, RMSE, or MAPE score how close forecasts are to actual values.

Q396. Temporal leakage.

A. Temporal leakage occurs when future information sneaks into training, inflating results.

G6. Additional advanced topics

Q397. Robust statistics and resistant estimators.

A. Robust statistics uses resistant estimators that are less affected by outliers and assumption violations.

Q398. Nonlinear statistical modeling.

A. Nonlinear statistical modeling captures curved or flexible relationships.

Q399. Survival analysis and censoring.

A. Survival analysis models time until an event, handling censored observations where the event isn't seen.

Q400. Missing-data theory and multiple imputation.

A. Multiple imputation fills missing values several times to reflect uncertainty in the gaps.

Q401. Causal inference: potential outcomes and treatment effects.

A. Causal inference with potential outcomes estimates the effect of a treatment by comparing what happens with and without it.

Q402. Propensity scores and matching.

A. Propensity scores estimate each unit's probability of treatment, and matching pairs similar units to balance groups.

Q403. Instrumental variables and regression discontinuity (introductory).

A. Instrumental variables and regression discontinuity estimate causal effects using external variables or cutoffs.

Q404. Bootstrap and simulation-based inference.

A. Bootstrap and simulation-based inference quantify uncertainty without strong distributional assumptions.

Q405. Monte Carlo methods.

A. Monte Carlo methods use random draws to approximate quantities that are hard to compute exactly.

Q406. Statistical learning theory.

A. Statistical learning theory studies how well learning algorithms generalize to new data.

Q407. Information theory: entropy, cross-entropy, and information gain.

A. Entropy measures uncertainty in a distribution, and cross-entropy and information gain are used to train and split models.

Q408. Multiple hypothesis testing at scale.

A. Multiple hypothesis testing at scale controls false positives across very large numbers of tests.

Q409. Reproducibility and uncertainty reporting.

A. Reproducibility and uncertainty reporting means sharing data, code, and uncertainty so others can verify results.

Q410. Fairness and subgroup performance evaluation.

A. Fairness and subgroup performance evaluation checks whether a model performs equally well across different groups.